

[2026-05-01T08:52:20.487486] Model: openai/gpt-5.2 === SYSTEM PROMPT === You are an expert scientific systematic reviewer specializing in the critical appraisal of academic literature for risk of bias. Your task is to evaluate the provided study objectively and rigorously. Please answer the evaluation question based STRICTLY on the text of the paper provided below. Do not use outside knowledge.

Instructions for Evaluation:

1. Evidence Extraction: First, locate and quote the exact, relevant passages from the paper.
2. Reasoning: Explain step-by-step how the quoted evidence answers the specific question. Do not assume or guess; if the information is missing, note that it is missing.
3. Final Judgment: Based strictly on the reasoning, select the most appropriate option.

You MUST provide ALL THREE sections above. Do NOT omit reasoning.

Definitions for ALL Valid Options (not all of the below answers will be valid for all questions. Please be aware of this):

- 'Yes' / 'No': The paper explicitly states information confirming the answer.
- 'Seemingly Yes' / 'Seemingly No': The paper strongly implies the answer, but does not explicitly state it.
- 'Unclear': The paper does not provide enough details to make a judgment.

You MUST conclude your response with the final selection in double brackets. Example format: [[Seemingly Yes]]

--- PAPER TEXT --- Plant Ecol (2009) 203:59–67 DOI 10.1007/s11258-008-9509-8

Deposition of exotic bird-dispersed seeds into three habitats of a fragmented landscape in the northeastern United States

Timothy S. McCay · Deanna H. McCay · Jessica L. Czajka

Received: 15 May 2008/Accepted: 12 September 2008/Published online: 30 September 2008 © Springer Science+Business Media B.V. 2008

Abstract Dispersal by frugivorous birds facilitates invasion by many exotic plants. We measured the seed rain of ornithochorous native and exotic plants at three habitats of a fragmented landscape of the northeastern United States for 1 year. We studied maple-beech forests, old fields, and abandoned conifer plantations. Across all sites we collected 2,196 ornithochorous seeds, including seeds from six exotic species and 10 native species. The majority (90%) of collected seeds were from exotic species. Seed dispersal was broadly similar among habitats, though seed rain of exotic species was higher in old

was broadly similar among habitats, though seed rain of exotic species was higher in old fields than forested habitats. Seed rain was not strongly influenced by artificial perches for most species. However, seeds of exotic species were more commonly found in traps under an artificial perch in old fields. Seed rains for the exotic *Elaeagnus umbellata*, *Rhamnus cathartica*, and *Rosa multiflora* were positively associated with local density of mature plants. Seed rain of *R. cathartica* was positively associated with abundance of seedlings but not saplings, suggesting that post-dispersal mortality was important. Seed dispersal of the exotic *Lonicera* spp. was

high in all habitats, accounting for 66% of all seeds collected. With the exception of *Lonicera* spp., seed rain of common exotic invaders was affected by the abundance of seed sources, and these species might be effectively controlled by elimination of local fruiting plants. Fruits of *Lonicera morrowii*, which has extensively invaded our area, are apparently a common component in the diet of frugivores.

Keywords Exotic species · Fragmented forest landscape · Invasion · Old fields · Plantations · Seed dispersal

Introduction

Colonization of new sites by invasive plants depends on dispersal of propagules and successful germination, growth, and reproduction. Although suitability of microsites clearly is important, propagule pressure may be a factor of primary importance in certain plant invasions (Rouget and Richardson 2003; Lambrinos 2006; DiVittorio et al. 2007; Milton et al. 2007). Exotic plants that produce fleshy fruits and are dispersed by animal vectors likely exist in the absence of historical dispersal agents within their new range (but see Simberloff and Von Holle 1999; Richardson et al. 2000). However, frugivorous vertebrates typically have catholic diets, and coevolved dispersal systems are generally flexible (Wheelwright and Orians 1982; Herrera 1985; Jordano 1995; Witmer

T. S. McCay (✉) · J. L. Czajka Department of Biology, Colgate University, Hamilton, NY 13346, USA e-mail: tmccay@mail.colgate.edu

D. H. McCay Department of Geography, Colgate University, Hamilton, NY 13346, USA

Springer

1996; Renne et al. 2002). Aronson et al. (2007) found that woody exotic plants with fleshy fruits were more successful in increasing their distributions than other exotic and native plants in the New York metropolitan region, suggesting that frugivores are frequently unwitting accomplices in the expansion of invaders.

Northeastern North America is inhabited by a suite of bird-dispersed (ornithochorous) exotic shrubs, including members of *Berberis*, *Elaeagnus*, *Lonicera*, *Rhamnus*, and *Rosa*. White and Stiles (1992) found that about one-third of the plant species dispersed by frugivorous birds in central New Jersey were exotic. *Lonicera maackii* (Amur honeysuckle)

is successfully dispersed by the American robin (*Turdus migratorius*), hermit thrush (*Catharus guttatus*), northern mockingbird (*Mimus polyglottos*), European starling (*Sturnus vulgaris*), and cedar waxwing (*Bombycilla cedrorum*) in Ohio (Bartuszevige and Gorchoy 2006); the northern mockingbird is a dispersal agent for *Rosa multiflora* (multiflora rose; Stiles 1982); the American robin readily consumes fruits of the invasive *Rhamnus frangula* (glossy buckthorn; LaFleur et al. 2007); and cedar waxwings (*Bombycilla cedrorum*) have incorporated *Berberis vulgaris* and *Rhamnus cathartica* into their diet (Witmer 1996). Birds of eastern North America clearly illustrate the common flexibility within coevolved avian--plant dispersal systems. The European starling (*Sturnus vulgaris*) is an effective frugivore that has become naturalized and common in much of North America. It consumes the fruits of a variety of native and exotic plants, including some occurring within its native range, in North America (McIlhenny 1936; Lindsey 1939; Russell 1971).

Frugivorous birds are more likely to defecate or regurgitate seeds near perches (McDonnell and Stiles 1983; Holl 1998; Shiels and Walker 2003). This tendency makes certain microsites—such as under shrubs, fencerows, and power lines—more likely to receive bird-dispersed seeds (Glyphis et al. 1981; Dean and Milton 2000). Perch use by frugivorous birds affects the rate and spatial pattern of encroachment by woody plants into old fields (Yarranton and Morrison 1974; McDonnell and Stiles 1983) and may affect the invasibility of different ecosystems by ornithochorous exotic plants (Milton et al. 2007; Deckers et al. 2008).

The landscape of central New York State is typical of much of eastern North America, including second-growth forest interspersed with agricultural land, abandoned agricultural land (old fields), plantations of trees managed for timber, and abandoned plantations. Areas recovering from disturbance, including old fields and abandoned plantations, may be more susceptible to invasion by exotics than more mature ecosystems (Fox and Fox 1986; Huebner and Tobin 2006). We measured the rate at which seeds of fleshy-fruited plants entered old fields, abandoned conifer plantations, and second-growth maple-beech forests of central New York State by bird dispersal in the presence and absence of artificial perches. We also related seed rain to the abundance of common exotics in these habitats. This study was part of a larger effort to understand invasion by *R. cathartica* (European buckthorn) at these sites (T.S. McCay, unpubl. results). Thus, particular emphasis was placed on understanding the role of propagule pressure in facilitating invasion by this species.

Methods

Study area

Our study area was in Madison County, near the geometric center of New York State, on properties owned by Colgate University (47°39'N, 75°35'W). Twelve plots were established, with one plot in each of maple-beech forest, old field, and conifer plantation

within each of four study areas: natural areas adjacent to the Colgate University campus (Campus), the Parker Farm natural area (Parker Farm), the southern end of the Bewkes Nature Preserve (Bewkes South), and the northern end of the Bewkes Nature Preserve (Bewkes North). Density of exotic species at these plots was typical of similar habitats in our landscape. Plots at northern hardwood forests and old fields were 30 × 50 m; plots at plantations were 30 × 30 m. Unmanageably large densities of *R. cathartica*, which were completely censused, precluded the use of 30 × 50-m plots in plantations.

Maple-beech forest stands were 60--120 years old based on increment cores of the largest individuals in the plots. Common overstory trees included *Acer saccharum* (sugar maple), *Fagus grandifolia* (American beech), and *Fraxinus americanus* (white ash). Herbaceous cover was sparse ($18 \pm 12\%$ [SD]) and shrubs were uncommon in maple-beech forests. Plantations were 35--45 years old and had not been

thinned since aforestation. Plantations were *Larix decidua* (European larch; Bewkes South and Bewkes North) or a mix of *Pinus resinosa* (red pine) and *Picea glauca* (white spruce; Campus and Parker Farm). High mortality of overstory trees resulted in canopy-gap formation and higher herbaceous cover ($57 \pm 21\%$) than in maple-beech forests. Plantation abandonment is common in our area, and these plantation species are not regenerating within stands (Goldblum 1998). Shrubs, including *L. morrowii* and *R. cathartica*, were common in plantations (shrub density $0.08 \pm 0.07 \text{ m}^{-2}$). Old fields were abandoned from cultivation or grazing 10--30 years ago based on age of woody vegetation. Old fields were dominated by grasses and other herbaceous plants ($86 \pm 11\%$ herbaceous cover); however, *L. morrowii*, *R. cathartica*, *F. americanus*, and *L. decidua* were common woody colonizers (shrub density $0.05 \pm 0.03 \text{ m}^{-2}$, tree density $0.03 \pm 0.04 \text{ m}^{-2}$; species classified as shrubs and trees based on growth form).

Sampling of seed rain

Ten 1-m^2 seed traps were established at random locations within each site. Traps consisted of a wooden frame made of untreated hemlock and a bottom made of nylon screen supported by 0.6-cm mesh galvanized steel hardware cloth. Traps were elevated 25 cm from the ground with wooden stakes. One half of the traps at each plot (5) included a bird perch emerging from their center, modified from the "horizontal" design of McDonnell and Stiles (1983). Perches consisted of 3 × 3-cm pressure-treated pine, were 2 m tall, and were topped with a single 1-m horizontal cross bar.

Seeds were recovered from traps approximately every 2 months between June 2, 2003 and June 15, 2004. Seeds were identified using Martin and Barkley (1961), and Young and Young (1992). Only seeds of ornithochorous plants were considered in the current analysis. We only considered seeds that were free of intact fruits and that were likely dispersed by birds (secondary seed rain, Kollmann and Pirl 1995). Thus, we disregarded entire fruits that fell into traps. We also disregarded a single large deposit of seeds that resulted from raccoon (*Procyon lotor*) defecation. We did not attempt sub-generic identification of seeds from the genera *Vitis*, *Lonicera*, and *Rubus* because of similarity in

seed morphology among congeneric species. The only species of *Lonicera* growing at our plots was *L. morrowii*. Plants were divided into exotic and native categories following Mitchell and Tucker (1997).

Sampling of vegetation

Plots were systematically searched during the summers of 2001 (Parker Farm), 2002 (Campus and Bewkes South), and 2003 (Bewkes North). All individuals of *R. cathartica* were tallied and measured for basal diameter pursuant to other aims of the work at these plots. Buckthorn individuals were separated into seedling (≤ 5 mm basal diameter [BD]), sapling ($0.5 < \text{BD} < 1.5$ cm), and adult (≥ 1.5 cm BD) life history stages. The threshold separating adults and saplings was based on the observation that the smallest *R. cathartica* to reproduce were 1.5 cm BD. The threshold separating seedlings and saplings reflected what we perceived to be a natural break in the distribution of plant sizes. All other plants were tallied if they were > 1 m tall.

Analysis

Data were only considered for those species for which we recovered ≥ 3 total seeds over all plots. We compared numbers of seeds for exotic and native species among habitat types and between traps with perches and traps without perches using analysis of variance (ANOVA) after square root transformation ($\sqrt{\text{count} + 0.5}$). This transformation effectively normalizes variables that are expected to be distributed as a Poisson random variable and is common for count data (Sokal and Rohlf 1995). When differences among habitats were found, we evaluated specific contrasts using the Tukey method of multiple comparisons (T-method, Sokal and Rohlf 1995). Analyses were conducted using SPSS 14.0[®]. We did not conduct hypothesis tests for individual species to avoid inflation of experiment-wide error rate (Sokal and Rohlf 1995; García 2004).

We related seed rain of our most common exotic taxa to the abundance of those taxa, and life-history stages for *R. cathartica*, using ranged major axis (RMA) regression. RMA regression is a type of model II regression appropriate when (1) both variables include random error variance, (2) variables are not in the same physical units, and (3) it can be reasonably assumed that the error variance is proportional to the

Plant Ecol (2009) 203:59–67

variance of the variable (Legendre and Legendre 1998). RMA regression is similar to standard major axis regression (Sokal and Rohlf 1995), except that RMA allows testing of the null hypothesis that the slope = 0. The data were inspected for outliers prior to analysis and log-transformed to meet statistical assumptions. These analyses were conducted using Model II, a program by Legendre (2001).

RESULTS

We collected and identified 2,196 seeds representing 10 native and six exotic taxa of ornithochorous

plants, along with *Rubus*, which might have included native or exotic species in our area (Table 1). We collected an additional 12,596 seeds of non-ornithochorous plants in the course of this study, primarily reflecting dominant wind-dispersed plants in each habitat (e.g., *Acer saccharum* in maple-beech forests). Thus, seeds of ornithochorous plants made up approximately 15% of the numerical seed rain measured at these plots.

Collection totals of all native seeds combined did not differ among habitats or between traps with perches and those without perches ($P > 0.05$, Table 1). *Sambucus racemosa* was collected at all four plantation plots but no maple-beech or old field

Table 1 Total seeds collected at 10 1-m² seed traps in each of four plots in each of three habitats in Central New York between June 2003 and June 2004

Species	Maple-beech		Old field		Plantation		$F^{\{a\}}_{\{H\}}$
	Perch	No perch	Perch	No perch	Perch	No perch	
Native taxa							
<i>Actaea pachypoda</i> Elliot	0	2	0	0	1	0	
<i>Cornus amomum</i> Mill.	0	7	0	6	1	1	
<i>C. racemosa</i> Lam.	1	2	0	0	0	1	
<i>Parthenocissus quinquefolia</i> (L.) Planch.	4	0	0	0	0	0	
<i>Prunus serotina</i> Ehrh.	3	6	0	0	2	1	
<i>Prunus virginiana</i> L.	6	6	9	0	1	1	
<i>Rhus typhina</i> L.	0	9	1	0	11	15	
<i>Sambucus</i>							

<i>Sambucus racemosa</i> L.	0	0	0	0	8	5	
<i>Viburnum lentago</i> L.	2	0	0	0	1	0	
<i>Vitis</i> L.	1	15	10	6	7	2	
Total native species	17	47	20	12	32	26	1.76
Exotic species							
<i>Berberis thunbergii</i> DC.	9	1	6	3	57	48	
<i>Elaeagnus umbellata</i> Thunb.	1	2	69	105	8	23	
<i>Lonicera</i> L.	22	101	986	209	55	86	
<i>Rhamnus cathartica</i> L.	3	0	39	12	14	4	
<i>Rosa multiflora</i> Thunb.	3	2	85	18	4	2	
<i>Solanum dulcamara</i> L.	0	0	9	1	0	2	
Total exotic species	38	106	1,194	348	138	165	13.79**
Unknown							
<i>Rubus</i> L.	0	0	48	0	4	1	
Total	55	153	1,262	360	174	192	

One half of seed traps were provisioned with an artificial perch at each plot. Species were classified as native or exotic following Mitchell and Tucker (1997). Seed of *Rubus* spp., which could not be identified to species, might have represented native or exotic taxa. Analysis of variance was conducted after square-root transformation ($\sqrt{y + 0.5}$) a Test statistics provided for hypotheses of no effect of habitat, perch, and habitat $\times$ perch interaction. Asterisks indicate rejection of the null hypothesis of no effect at $\alpha = 0.05$: * indicates $P \leq 0.05$; ** indicates $P \leq 0.001$

Springer

Plant Ecol (2009) 203:59–67

Fig. 1 Relationships between seed rain and plant (≥ 1 m tall) density (both transformed by taking the natural logarithm of the variable after adding one) at 4 plots in each of maple-beech forests, old fields, and conifer plantations for *Elaeagnus umbellata*, *Lonicera morrowii*, and *Rosa multiflora*. Seeds were collected between June 2003 and June 2004. Significant relationships fit using ranged major axis regression are indicated with dashed lines

plots. Four of 10 native species, including *Actaea pachypoda*, *Cornus racemosa*, *Prunus serotina*, and *Viburnum lentago*, were not collected in old fields but were collected in maple-beech forest. *Parthenocissus quinquefolia* was only collected at maple-beech forest plots.

More seeds of exotic plants were collected in old fields than the other habitats (FHabitat = 13.79, df = 2,18, $P < 0.01$; Tukey Method $P < 0.05$), and perches were positively associated with seed rain at old fields but not other habitats (FH $\times$ P = 3.56, df = 2,18, $P = 0.05$). These trends were strongly influenced by the most common exotic taxon, *Lonicera* spp., which accounted for 73% of exotic seeds. Seed rain of *Lonicera* spp. was markedly higher at old fields and was apparently influenced by perches in old fields but not other habitats. Seeds of *Elaeagnus umbellata*, *Rhamnus cathartica*, and *Rosa multiflora* also were markedly more common at old fields than in forested habitats. Seeds of *B. thunbergii* were more commonly collected in plantations than other habitats.

Annual seed rain of *Elaeagnus* and *Rosa* each were explained well by the abundance of those taxa at our plots (*Elaeagnus*: slope = 0.84, $P < 0.01$; *Rosa*: slope = 0.89, $P < 0.01$; Fig. 1). Seed rain of *R. cathartica* was positively related to the density of *R. cathartica* seedlings (slope = 0.70, $P = 0.01$) and adults (slope = 0.79, $P = 0.03$) but not saplings ($P = 0.32$) at our plots (Fig. 2). Seed rain of *Lonicera* was consistently high among all plots regardless of the density of large (> 1 m tall) *L. morrowii* (Fig. 1), which varied markedly among maple-beech forests (median = 7 ha $^{-1}$, range = [0, 13]), old fields (150 ha $^{-1}$, [53, 467]), and plantations (378 ha $^{-1}$, [189, 1767]). *Lonicera morrowii* was the most widespread exotic shrub at our plots, detected at 11 of our 12 plots.

When species were combined within exotic and native categories, we found a much greater seed rain

Springer

Plant Ecol (2009) 203:59–67

Fig. 2 Relationships between seed rain and density of *Rhamnus cathartica* seedlings (< 5 mm basal diameter [BD]), saplings (5 mm $\leq$ BD < 15 mm), and adult plants (≥ 15 mm BD) at 12 plots in central New York. All variables were transformed by taking the natural logarithm of the variable after adding one. Seeds were collected between June 2003 and June 2004. Vegetation was sampled during summers of 2001–2003. Significant

relationships fit using ranged major axis regression are indicated with dashed lines

of exotic species than native species ($F_{\text{Origin}} = 45.08$, $df = 1,36$, $P < 0.01$, Fig. 3). Total seed captures also were affected by habitat ($F_{\text{Habitat}} = 10.20$, $df = 2,36$, $P < 0.01$), with more seeds collected at old fields than other habitats (Tukey Method, $P < 0.05$). The interaction between origin and habitat was significant ($F_{\text{O} \times \text{H}} = 15.41$, $df = 2,36$, $P < 0.01$), with relatively more exotic seeds collected in old fields than other habitats (Fig. 3). Finally, artificial perches influenced total seed capture positively in old fields but negligibly in other habitats ($F_{\text{P} \times \text{H}} = 4.08$, $df = 2,36$, $P = 0.03$; Fig. 4). These trends were all strongly influenced by seed rain of *Lonicera* spp., which accounted for 66% of total seeds.

Discussion

Rejmanek (1996) identified effective vertebrate seed dispersal as a factor associated with invasion success by exotic seed plants. We found that several of the most common woody exotic invaders of our landscape are dispersed by local avian frugivores. Thirty-eight percent (6/16) of ornithochorous plant species recovered in our seed traps were exotic to our area. Similarly, White and Stiles (1992) found that one-third of ornithochorous species collected in seed traps in central New Jersey were exotic. A large fraction of seeds (90%) dispersed by birds in our area were from exotic plants, particularly *Lonicera* spp. White and Stiles (1992) estimated that fruits of exotic plants made up $\leq 14\%$ of estimated fruit biomass consumed by birds in central New Jersey during 1977–1986.

We found greater seed rain, particularly for exotic plants, in old fields than in forests. This is likely due to the patterns of habitat use by birds. Seed rain of *E. umbellata*, *R. cathartica*, and *R. multiflora* were

Springer

Plant Ecol (2009) 203:59–67

Fig. 3 Mean seed rain ($\pm$ s.e.) of exotic and native (to New York State) ornithochorous plants in maple-beech forests, old fields, and conifer plantations in central New York. Collections were made over 1 year from June 2003 to June 2004. The scale of the y-axis is logarithmic and begins at 0.1

Fig. 4 Mean seed rain ($\pm$ s.e.) of ornithochorous plants into maple-beech forests, old fields, and conifer plantations based on collections made using seed traps including an artificial perch and those not including an artificial perch over 1 year. The scale of the y-axis is logarithmic and begins at 0.1

explained well by the abundance of mature plants at our plots. Cause and effect might possibly go both ways in these relationships, with areas of intense seed rain leading to

large adult populations or areas with many adult plants causing high seed rain. Seed rain of *R. cathartica* was related to seedling density, suggesting that high seed rain might lead to high seedling densities. However, seed rain of *R. cathartica* was not related to sapling density. In a population of *Olea europaea* the spatial distributions of seedlings and mature plants differed, suggesting non-random post-dispersal survivorship (Rey and Alcantara 2000). Differences in distribution between seedlings and saplings of the invasive *Prunus serotina* in Belgium may have been related to differential response to light

(Godefroid et al. 2005). Similarly, we suggest that conditions promoting dispersal and germination of *R. cathartica*, including proximity to shrubs, may increase mortality of seedlings and reduce likelihood of maturation.

Local density of adult plants, which are potential seed sources, did not affect propagule pressure of *Lonicera*. Mature $>1\mathrm{m}$ *L. morrowii* were very common, found at 11 of our 12 plots. It may be that *Lonicera* fruits make up a significant portion of the diets of birds in our area, and dispersal is generally high in all habitats. Fruits of *L. morrowii* are sugary and may provide an important energetic resource for certain frugivores during the autumn and winter (Witmer and Van Soest 1998), as the fruits of the invasive *L. tatarica* do in other areas of North America (Drummond 2005).

Unlike many authors working in early-successional and disturbed systems (McDonnell and Stiles 1983; Holl 1998; Shiels and Walker 2003), we found that perch presence had a small effect on seed rain relative to large spatial variability for most species. The perches that we used were similar to the cross-bar type artificial perch used by Holl (1998), which was less effective than natural perches constructed of dead tree branches. However, McDonnell and Stiles (1983) found similar rates of seed deposition under natural and cross-bar type artificial perches. Our perches may have generally increased seed rain across the entire plot, thus being reflected in both traps with artificial perches and those without artificial perches. McDonnell and Stiles (1983) found that the introduction of perches generally elevated rates of dispersal in large areas surrounding the perch. Our traps without perches also may have attracted birds (Kollmann and Goetze 1998). Most investigators noting an increase in dispersal with artificial perches worked in areas with few natural perches (McDonnell and Stiles 1983; Holl 1998). It is noteworthy that our perches seem to have had a greater positive effect on seed capture in old fields than other habitats. Natural perches were more available at forested plots, potentially freeing birds from the use of artificial perches.

Our results did not document the identity of dispersal agents or the success of dispersal, including subsequent germination. Seeds of the invasive *Lonicera maackii* were not inhibited by passage by American robins but were inhibited by cedar waxwings (Bartuszevige and Gorchov 2006). There may

Springer

Plant Ecol (2009) 203:59–67

be important associations between identity and habitat use of dispersal agents and the survival of seeds of particular exotic plants in our landscape.

The remote location of our plots and large number of traps used in the study prevented us from frequent collection of seeds. The time between visits to traps may have allowed granivores to predate seeds, which would cause us to underestimate seed rain. Kollmann and Goetze (1998) summarized the strong evidence that seed predation can markedly influence seed-capture rates and that seed predation can vary among habitats. In our case, this may have exaggerated small seed-capture rates in maple-beech forests because mammalian granivores likely were most common in that habitat. In contrast, Stevenson and Vargas (2008) found seed predation rates lower in seed traps than at the nearby forest floor in tropical forests. Unfortunately we did not quantify rates of seed loss with labeled seeds. Time between visits also may have allowed fruits to degrade and mammal feces to disintegrate, especially during warm months. It is therefore possible that some of the seed deposition that we attribute to secondary seed rain by birds is primary seed rain or secondary seed rain by mammals.

The most common types of ornithochorous seeds dispersed to all ecosystems in central New York were of invasive, exotic plants. Seed rain of *E. umbellata*, *R. multiflora*, and *R. cathartica* reflected the local density of adult plants that might serve as sources of propagules. Managers aiming to control the spread of these species in our area might, therefore, consider selective removal of propagule sources (Gosper et al. 2005). The seed rain of *L. morrowii* was generally high regardless of the local density of seed sources. However, the abundance of mature *L. morrowii* was much greater in old fields and plantations than maple-beech forests, possibly reflecting greater post-dispersal mortality in mature forests. The control of *Lonicera* might therefore involve physical removal or management for mature, closed-canopy habitats, which are poorly invaded by members of this genus (Robertson et al. 1994; Hutchinson and Vankat 1997).

Acknowledgments The authors wish to thank AV Caragiulo and TL Mandel for their help in the field and E Juers for help in the lab. Comments by two anonymous reviewers greatly improved an earlier draft of the manuscript. This paper is based upon work supported by the National Science Foundation under Grant No. BCS-0111743. This work was supported, in part, by the Colgate University Research Council.

References

- Aronson HFJ, Handel SN, Clemants SE (2007) Fruit type, life form and origin determine the success of woody plant invaders in an urban landscape. *Biol Invasions* 9:465-475. doi:10.1007/s10530-006-9053-1
- Bartuszevige AM, Gorchov DL (2006) Avian seed dispersal of an invasive shrub. *Biol Invasions* 8:1013-1022. doi: 10.1007/s10530-005-3634-2
- Dean WRJ, Milton SJ (2000) Directed dispersal of *Opuntia* species in the Karoo, South Africa: are crows the responsible agents? *J Arid Environ* 15:305-311. doi:10.1006/jare.2000.0652
- Deckers B, Verbeyen K, Vanhellemont M, Maddens E, Muys B, Hermv M (2008) Impact of avian frugivores on dispersal and recruitment of the invasive

Merriam W (2000) Impact of avian frugivores on dispersal and recruitment of the invasive *Prunus serotina* in an agricultural landscape. *Biol Invasions* 10:717–727. doi: 10.1007/s10530-007-9164-3 DiVittorio CT, Corbin JD, D'Antonio CM (2007) Spatial and temporal patterns of seed dispersal: an important determinant of grassland invasions. *Ecol Appl* 17:311–316. doi:10.1890/06-0610 Drummond BA (2005) The selection of native and invasive plants by frugivorous birds in Maine. *Northeast Nat* 12: 33–44. doi:10.1656/1092-6194(2005)012[0033: TSONAI]2.0.CO;2 Fox MD, Fox BJ (1986) The susceptibility of natural communities to invasion. In: Groves RH, Burdon JJ (eds) *Ecology of biological invasions*. Cambridge University Press, Cambridge, UK, pp 57–66 García LV (2004) Escaping the Bonferroni iron claw in ecological studies. *Oikos* 105:657–663. doi:10.1111/j.0030-1299.2004.13046.x Glyphis JP, Milton SF, Siegfried WR (1981) Dispersal of *Acacia cyclops* by birds. *Oecologia* 48:138–141. doi: 10.1007/BF00347002 Godefroid S, Phartyal SS, Weyembergh G, Koedam N (2005) Ecological factors controlling the abundance of non-native invasive black cherry (*Prunus serotina*) in deciduous forest understory in Belgium. *For Ecol Manage* 210: 91–105 Goldblum D (1998) Regeneration in unmanaged conifer plantations, upstate New York. *Northeast Nat* 5:343–358. doi:10.2307/3858565 Gosper CR, Stansbury CD, Vivian-Smith G (2005) Seed dispersal of fleshy-fruited invasive plants by birds: contributing factors and management options. *Divers Distrib* 11:549–558. doi:10.1111/j.1366-9516.2005.00195.x Herrera CM (1985) Determinants of plant-animal coevolution: the case of mutualistic dispersal of seeds by vertebrates. *Oikos* 44:132–141. doi:10.2307/3544054 Holl KD (1998) Do bird perching structures elevate seed rain and seedling establishment in abandoned tropical pasture? *Restor Ecol* 6:253–261. doi:10.1046/j.1526-100X.1998.00638.x

Springer

Plant Ecol (2009) 203:59–67

Huebner CD, Tobin PC (2006) Invasibility of mature and 15-year-old deciduous forests by exotic plants. *Plant Ecol* 186:57–68. doi:10.1007/s11258-006-9112-9 Hutchinson TF, Vankat JL (1997) Invasibility and effects of Amur honeysuckle in Southwestern Ohio forests. *Conserv Biol* 11:1117–1124. doi:10.1046/j.1523-1739.1997.96001.x Jordano P (1995) Angiosperm fleshy fruits and seed dispersers: a comparative analysis of adaptation and constraints in plant-animal interactions. *Am Nat* 145:163–191. doi: 10.1086/285735 Kollmann J, Goetze D (1998) Notes on seed traps in terrestrial plant communities. *Flora* 193:31–40 Kollmann J, Pirl M (1995) Spatial pattern of seed rain of fleshy-fruited plants in a scrubland-grassland transition. *Acta Oecol* 16:313–329 LaFleur NE, Rubega MA, Elphick CS (2007) Invasive fruits, novel foods, and choice: an investigation of European starling and American robin frugivory. *Wilson J Ornithol* 119:429–438. doi:10.1676/05-115.1 Lambrinos JG (2006) Spatially variable propagule pressure and herbivory influence invasion of chaparral shrubland by an exotic grass. *Oecologia* 147:327–334. doi:10.1007/s00442-005-0259-1 Legendre P (2001) *Model II regression—user's guide*. Département de Sciences Biologiques, Université de Montréal Legendre P, Legendre L (1998) *Numerical ecology*, 2nd edn. Elsevier Science, Amsterdam Lindsey AA (1939) Food of the starling in central New York State. *Wilson Bull* 51:176–182 Martin AC, Barkley WD

(1961) Seed identification manual. University of California Press, Berkeley McDonnell MJ, Stiles EW (1983) The structural complexity of old field vegetation and the recruitment of bird-dispersed plant species. *Oecologia* 56:109-116. doi:10.1007/BF00378225 McIlhenny EA (1936) Are starlings a menace to the food supply of our native birds? *Auk* 53:338-339 Milton SJ, Wilson JR, Richardson DM, Seymour CL, Dean WRJ, Iponga DM, Proches S (2007) Invasive alien plants infiltrate bird-mediated shrub nucleation processes in arid savanna. *J Ecol* 95:648-661. doi:10.1111/j.1365-2745.2007.01247.x Mitchell RS, Tucker GC (1997) Revised checklist of New York State plants. Bulletin No 490. The New York State Museum, Albany, New York Rejmánek M (1996) A theory of seed plant invasiveness: The first sketch. *Biol Conserv* 78:171-181. doi:10.1016/0006-3207(96)00026-2 Renne IJ, Barrow WC Jr, Johnson Randall LA, Bridges WC Jr (2002) Generalized avian dispersal syndrome contributes to Chinese tallow tree (*Sapium sebiferum*, Euphorbiaceae) invasiveness. *Divers Distrib* 8:285-295. doi:10.1046/j.1472-4642.2002.00150.x

Rey PJ, Alcantara JM (2000) Recruitment dynamics of a fleshy-fruited plant (*Olea europaea*): connecting patterns of seed dispersal to seedling establishment. *J Ecol* 88:622-633. doi:10.1046/j.1365-2745.2000.00472.x Richardson DM, Allsopp D, D'Antonio CM, Milton SJ, Rejmánek M (2000) Plant invasions-the role of mutualisms. *Biol Rev Camb Philos Soc* 75:65-93. doi:10.1017/S0006323199005435 Robertson DJ, Robertson MC, Tague T (1994) Colonization dynamics of four exotic plants in a northern Piedmont natural area. *J Torrey Bot Soc* 121:107-118 Rouget M, Richardson DM (2003) Inferring process from pattern in plant invasions: a semimechanistic model incorporating propagule pressure and environmental factors. *Am Nat* 162:713-724. doi:10.1086/379204 Russell DN (1971) Food habits of the starling in eastern Texas. *Condor* 73:369-372. doi:10.2307/1365766 Shiels AB, Walker LR (2003) Bird perches increase forest seeds on Puerto Rican landslides. *Restor Ecol* 11:457-465. doi:10.1046/j.1526-100X.2003.recl0269.x Simberloff D, Von Holle B (1999) Positive interactions of nonindigenous species: invasional meltdown? *Biol Invasions* 1:21-32. doi:10.1023/A:1010086329619 Sokal RR, Rohlf FJ (1995) Biometry: the principles and practice of statistics in biological research, 3rd edn. W. H. Freeman and Company, New York Stevenson PR, Vargas IN (2008) Sample size and appropriate design of fruit and seed traps in tropical forests. *J Trop Ecol* 24:95-105. doi:10.1017/S0266467407004646 Stiles EW (1982) Expansions of mockingbird and multiflora rose in the northeastern United States and Canada. *Am Birds* 36:358-364 Wheelwright NT, Orians GH (1982) Seed dispersal of plants by animals: contrasts with pollen dispersal, problems of terminology, and constraints on evolution. *Am Nat* 119:402-413. doi:10.1086/283918 White DW, Stiles EW (1992) Bird dispersal of fruits of species introduced into eastern North America. *Can J Bot* 70:1689-1696 Witmer MC (1996) Annual diet of cedar waxwings based on U.S. Biological Survey records (1885-1950) compared to diet of American robins: contrasts in dietary patterns and natural history. *Auk* 113:414-430 Witmer MC, Van Soest PJ (1998) Contrasting digestive strategies of fruit-eating birds. *Funct Ecol* 12:728-741. doi:10.1046/j.1365-2435.1998.00242.x Yarranton GA, Morrison RG (1974) Spatial dynamics of a primary succession: nucleation. *J Ecol* 62:417-428. doi: 10.2307/2258988 Young JA, Young CG (1992) Seeds of woody plants in North America. Dioscorides Press, Oregon

=== USER PROMPT === Question: 1.1. Is it possible for the impact of the exposure or the effectiveness of the intervention to be confounded in this study?

Important: Your final response MUST be exactly one of the following valid options: 'yes', 'seemingly yes', 'seemingly no', 'no'. You MUST conclude your response with the final selection enclosed in double brackets, e.g., [[Yes]]. === RAW RESPONSE ===

Evidence extraction (exact quotes)

- Observational comparison across habitats (non-randomized exposure):
 - "We studied maple-beech forests, old fields, and abandoned conifer plantations."
 - "Twelve plots were established, with one plot in each of maple-beech forest, old field, and conifer plantation within each of four study areas..."
- Non-experimental association of exposure with outcomes (propagule pressure vs local abundance), with authors acknowledging non-causal direction:
 - "We also related seed rain to the abundance of common exotics in these habitats."
 - "Cause and effect might possibly go both ways in these relationships, with areas of intense seed rain leading to large adult populations or areas with many adult plants causing high seed rain."
- Differences among habitats that could co-vary with seed rain (potential confounders):
 - "Old fields were dominated by grasses and other herbaceous plants ($86 \pm 11\%$ herbaceous cover)..."
 - "Herbaceous cover was sparse ($18 \pm 12\%$ [SD]) and shrubs were uncommon in maple-beech forests."
 - "Shrubs, including *L. morrowii* and *R. cathartica*, were common in plantations..."

Reasoning (step-by-step)

1. The main "exposures" in the study include habitat type (maple-beech forest vs old field vs plantation) and local abundance/density of exotic plants (as seed sources), and the outcomes are measures of "seed rain" into traps.
2. Habitat type is not randomly assigned; the study compares existing habitat categories ("We studied..."; "plots were established..."). Because habitats differ systematically in vegetation structure and shrub abundance (quotes above), these habitat-linked factors can influence bird behavior/perching and seed deposition, and they also differ by habitat.
3. The study explicitly relates seed rain to the abundance of mature plants ("related seed

rain to the abundance..."). The authors acknowledge ambiguity in causal direction ("Cause and effect might possibly go both ways..."), indicating that observed associations may reflect other linked factors rather than a clean causal effect of the exposure.

4. Since these exposures are not controlled in a way that would block alternative explanations, it is possible that the estimated impact of the exposure/effectiveness (e.g., habitat differences or plant-source abundance effects) is confounded by other co-varying habitat and vegetation characteristics.

Final judgment

**Confounding is possible in this study. [[yes]] === PARSED
OUTPUT === yes**

[2026-05-01T08:52:38.120465] Model: openai/gpt-5.2 === SYSTEM PROMPT === You are an expert scientific systematic reviewer specializing in the critical appraisal of academic literature for risk of bias. Your task is to evaluate the provided study objectively and rigorously. Please answer the evaluation question based STRICTLY on the text of the paper provided below. Do not use outside knowledge.

Instructions for Evaluation:

1. Evidence Extraction: First, locate and quote the exact, relevant passages from the paper.
2. Reasoning: Explain step-by-step how the quoted evidence answers the specific question. Do not assume or guess; if the information is missing, note that it is missing.
3. Final Judgment: Based strictly on the reasoning, select the most appropriate option.

You MUST provide ALL THREE sections above. Do NOT omit reasoning.

Definitions for ALL Valid Options (not all of the below answers will be valid for all questions. Please be aware of this):

- 'Yes' / 'No': The paper explicitly states information confirming the answer.
- 'Seemingly Yes' / 'Seemingly No': The paper strongly implies the answer, but does not explicitly state it.
- 'Unclear': The paper does not provide enough details to make a judgment.

You MUST conclude your response with the final selection in double brackets. Example format: [[Seemingly Yes]]

=== USER PROMPT === Previous appraisal context is provided for continuity. Use it only as local context, and judge the current question from the paper text.

Previous node decisions:

- q_1_1: 1.1. Is it possible for the impact of the exposure or the effectiveness of the intervention to be confounded in this study? Answer: yes

Question: 1.2. Did the author(s) control for all the potential confounders?

Important: Your final response MUST be exactly one of the following valid options: 'yes', 'seemingly yes', 'seemingly no', 'no', 'unclear'. You MUST conclude your response with the final selection enclosed in double brackets, e.g., [[Yes]]. === RAW RESPONSE ===

1) Evidence extraction (verbatim quotes)

- Design/controls included in analysis:
 - "We compared numbers of seeds for exotic and native species among habitat types and between traps with perches and traps without perches using analysis of variance (ANOVA)..."
 - "We related seed rain of our most common exotic taxa to the abundance of those taxa, and life-history stages for *R. cathartica*, using ranged major axis (RMA) regression."
- Explicitly stated important uncontrolled/uncertain sources of bias/confounding:
 - "The time between visits to traps may have allowed granivores to predate seeds, which would cause us to underestimate seed rain."
 - "Unfortunately we did not quantify rates of seed loss with labeled seeds."
 - "Time between visits also may have allowed fruits to degrade and mammal feces to disintegrate... It is therefore possible that some of the seed deposition that we attribute to secondary seed rain by birds is primary seed rain or secondary seed rain by mammals."
 - "Our results did not document the identity of dispersal agents or the success of dispersal, including subsequent germination."
- Indication that other habitat differences could affect the perch comparison (i.e., a plausible confounder not "controlled for" beyond stratifying by habitat):
 - "Natural perches were more available at forested plots, potentially freeing birds from the use of artificial perches."

2) Reasoning (step-by-step, based strictly on the paper text)

1. The authors *did* account for some key variables in their analyses—specifically habitat type and whether a trap had an artificial perch (ANOVA), and local plant abundance for certain taxa (RMA regressions). This indicates some attempt to address confounding by measured factors (e.g., habitat, seed source abundance).

2. However, the paper explicitly acknowledges multiple important factors that could affect measured seed rain and interpretation, and states they were **not** measured/quantified:
 - Seed predation in traps ("may have allowed granivores to predate seeds") with no quantification ("did not quantify rates of seed loss").
 - Misattribution of dispersal pathway due to trap visit interval (seed deposition attributed to birds might actually be from fruits degrading or mammal feces disintegrating).
 - Identity of dispersal agents and success of dispersal not documented.
3. Because the authors themselves identify these as limitations that could bias/alter observed seed rain patterns, and they did not measure/adjust for them, the study cannot be said (from the text) to have controlled for **all** potential confounders.

3) Final judgment

The authors controlled for some measured factors (habitat, perch presence, local plant abundance), but the paper explicitly states several plausible confounding/biasing factors were not measured or adjusted for (e.g., seed predation loss, uncertainty in dispersal source, unidentified dispersal agents). Therefore, they did **not** control for all potential confounders.

[[seemingly no]] === PARSED OUTPUT === seemingly no

[2026-05-01T08:52:51.822570] Model: openai/gpt-5.2 === SYSTEM PROMPT === You are an expert scientific systematic reviewer specializing in the critical appraisal of academic literature for risk of bias. Your task is to evaluate the provided study objectively and rigorously. Please answer the evaluation question based STRICTLY on the text of the paper provided below. Do not use outside knowledge.

Instructions for Evaluation:

1. Evidence Extraction: First, locate and quote the exact, relevant passages from the paper.
2. Reasoning: Explain step-by-step how the quoted evidence answers the specific question. Do not assume or guess; if the information is missing, note that it is missing.
3. Final Judgment: Based strictly on the reasoning, select the most appropriate option.

You MUST provide ALL THREE sections above. Do NOT omit reasoning.

Definitions for ALL Valid Options (not all of the below answers will be valid for all questions. Please be aware of this):

- 'Yes' / 'No': The paper explicitly states information confirming the answer.
- 'Seemingly Yes' / 'Seemingly No': The paper strongly implies the answer, but does not

seemingly yes / seemingly no'. The paper strongly implies the answer, but does not explicitly state it.

- 'Unclear': The paper does not provide enough details to make a judgment.

You MUST conclude your response with the final selection in double brackets. Example format: [[Seemingly Yes]]

--- PAPER TEXT ---

=== USER PROMPT === Previous appraisal context is provided for continuity. Use it only as local context, and judge the current question from the paper text.

Previous node decisions:

- q_1_1: 1.1. Is it possible for the impact of the exposure or the effectiveness of the intervention to be confounded in this study? Answer: yes
- q_1_2: 1.2. Did the author(s) control for all the potential confounders? Answer: seemingly no

Question: 1.3. Is there any justifiable reason for not controlling for all the potential confounders?

Important: Your final response MUST be exactly one of the following valid options: 'yes', 'seemingly yes', 'seemingly no', 'no'. You MUST conclude your response with the final selection enclosed in double brackets, e.g., [[Yes]]. === RAW RESPONSE ===

Evidence extraction (exact quotes)

1. Logistical constraints limiting measurement/control of a potential confounding process (seed predation/seed loss):

- **"The remote location of our plots and large number of traps used in the study prevented us from frequent collection of seeds. The time between visits to traps may have allowed granivores to predate seeds, which would cause us to underestimate seed rain."**
- **"Unfortunately we did not quantify rates of seed loss with labeled seeds."**

2. Additional acknowledgement that methodological constraints could affect attribution of dispersal agent (another potential source of bias/confounding in interpretation):

- **"Time between visits also may have allowed fruits to degrade and mammal feces to disintegrate, especially during warm months. It is therefore possible that some of the seed deposition that we attribute to secondary seed rain by birds is primary seed rain or secondary seed rain by mammals."**

Reasoning (step-by-step)

1. The question asks whether the paper provides **any justifiable reason** for not controlling for all potential confounders.
2. The authors explicitly describe a concrete, practical limitation: the "remote location" and "large number of traps" **prevented** frequent collection.
3. They explain that infrequent collection could allow "granivores to predate seeds," affecting measured seed rain (i.e., an unmeasured process that could differ across habitats and influence results, functioning as an uncontrolled confounding/bias source).
4. They further state they did not quantify seed loss with labeled seeds ("Unfortunately we did not quantify..."), directly indicating that this potential confounder/bias was not controlled due to the stated logistical constraints.
5. Because these passages provide an explicit logistical justification for not controlling/quantifying at least some potential confounding influences (e.g., seed predation/seed loss; possibly mammal vs bird deposition), the paper **does provide a justifiable reason**.

Final judgment

[[yes]] === PARSED OUTPUT === yes

[2026-05-01T08:53:04.521016] Model: openai/gpt-5.2 === SYSTEM PROMPT === You are an expert scientific systematic reviewer specializing in the critical appraisal of academic literature for risk of bias. Your task is to evaluate the provided study objectively and rigorously. Please answer the evaluation question based STRICTLY on the text of the paper provided below. Do not use outside knowledge.

Instructions for Evaluation:

1. Evidence Extraction: First, locate and quote the exact, relevant passages from the paper.
2. Reasoning: Explain step-by-step how the quoted evidence answers the specific question. Do not assume or guess; if the information is missing, note that it is missing.
3. Final Judgment: Based strictly on the reasoning, select the most appropriate option.

You MUST provide ALL THREE sections above. Do NOT omit reasoning.

Definitions for ALL Valid Options (not all of the below answers will be valid for all questions. Please be aware of this):

- 'Yes' / 'No': The paper explicitly states information confirming the answer.
- 'Seemingly Yes' / 'Seemingly No': The paper strongly implies the answer, but does not explicitly state it.
- 'Unclear': The paper does not provide enough details to make a judgment.

You MUST conclude your response with the final selection in double brackets. Example format: [[Seemingly Yes]]

--- PAPER TEXT ---

=== USER PROMPT === Previous appraisal context is provided for continuity. Use it only as local context, and judge the current question from the paper text.

Previous node decisions:

- q_1_2: 1.2. Did the author(s) control for all the potential confounders? Answer: seemingly no
- q_1_3: 1.3. Is there any justifiable reason for not controlling for all the potential confounders? Answer: yes

Question: 1.4. Were the potential confounders measured accurately and precisely enough?

Important: Your final response MUST be exactly one of the following valid options: 'yes', 'seemingly yes', 'seemingly no', 'no', 'unclear'. You MUST conclude your response with the final selection enclosed in double brackets, e.g., [[Yes]]. === RAW RESPONSE ===

Evidence Extraction (verbatim quotes)

- Vegetation measurement approach:

"Plots were systematically searched during the summers of 2001 (Parker Farm), 2002 (Campus and Bewkes South), and 2003 (Bewkes North). All individuals of *R. cathartica* were tallied and measured for basal diameter..."

"Buckthorn individuals were separated into seedling (≤ 5 mm basal diameter [BD]), sapling ($0.5 < BD < 1.5$ cm), and adult (≥ 1.5 cm BD)..."

"All other plants were tallied if they were > 1 m tall."

- Seed measurement/processing (relevant to exposure/outcome but also shows some limitations in classification):

"Seeds were recovered from traps approximately every 2 months..."

"Seeds were identified using Martin and Barkley (1961), and Young and Young (1992)."

"We did not attempt sub-generic identification of seeds from the genera *Vitis*, *Lonicera*, and *Rubus* because of similarity in seed morphology among congeneric species."

Reasoning (step-by-step, based strictly on the text)

1. The question asks whether *potential confounders* were measured "accurately and precisely enough." In this study, the main measured covariates that could act as confounders/effect modifiers in analyses include vegetation abundance/density (seed sources) and habitat (forest/old field/plantation), and perch presence.
2. The paper describes *how* vegetation was measured (systematic searches; tallying all *R. cathartica* individuals and measuring basal diameter; tallying all other plants only if >1 m tall). These passages demonstrate a defined measurement procedure and clear thresholds for *R. cathartica* life stages.
3. However, the paper does **not** report any validation, repeatability, inter-observer reliability, measurement error estimates, calibration procedures, or other information that would allow judging whether these confounder measurements were sufficiently accurate and precise.
4. Additionally, for some taxa the study explicitly limits taxonomic precision (e.g., not identifying *Lonicera* seeds to species level), but the question is about confounders; regardless, the text still does not quantify how accurate/precise plant abundance measures (or other confounders) are.
5. Therefore, from the paper text alone, there is not enough information to determine whether confounders were measured with adequate accuracy and precision.

Final Judgment

[[unclear]] === PARSED OUTPUT === unclear

[2026-05-01T08:53:23.564702] Model: openai/gpt-5.2 === SYSTEM PROMPT === You are an expert scientific systematic reviewer specializing in the critical appraisal of academic literature for risk of bias. Your task is to evaluate the provided study objectively and rigorously. Please answer the evaluation question based STRICTLY on the text of the paper provided below. Do not use outside knowledge.

Instructions for Evaluation:

1. Evidence Extraction: First, locate and quote the exact, relevant passages from the paper.
2. Reasoning: Explain step-by-step how the quoted evidence answers the specific question. Do not assume or guess; if the information is missing, note that it is missing.
3. Final Judgment: Based strictly on the reasoning, select the most appropriate option.

You MUST provide ALL THREE sections above. Do NOT omit reasoning.

Definitions for ALL Valid Options (not all of the below answers will be valid for all

Definitions for ALL valid options (not all of the below answers will be valid for all questions. Please be aware of this):

- 'Yes' / 'No': The paper explicitly states information confirming the answer.
- 'Seemingly Yes' / 'Seemingly No': The paper strongly implies the answer, but does not explicitly state it.
- 'Unclear': The paper does not provide enough details to make a judgment.

You MUST conclude your response with the final selection in double brackets. Example format: [[Seemingly Yes]]

--- PAPER TEXT ---

=== USER PROMPT === Previous appraisal context is provided for continuity. Use it only as local context, and judge the current question from the paper text.

Previous node decisions:

- q_1_3: 1.3. Is there any justifiable reason for not controlling for all the potential confounders? Answer: yes
- q_1_4_b: 1.4. Were the potential confounders measured accurately and precisely enough? Answer: unclear

Question: 1.5. Did the author(s) analyse the effect appropriately taking confounders into account?

Important: Your final response MUST be exactly one of the following valid options: 'yes', 'seemingly yes', 'seemingly no', 'no', 'unclear'. You MUST conclude your response with the final selection enclosed in double brackets, e.g., [[Yes]]. === RAW RESPONSE ===

Evidence extraction (verbatim quotes)

- On how effects were analysed (habitat/perch only):

"We compared numbers of seeds for exotic and native species among habitat types and between traps with perches and traps without perches using analysis of variance (ANOVA)..."

"When differences among habitats were found, we evaluated specific contrasts using the Tukey method of multiple comparisons..."

- On how seed rain–abundance relationships were analysed (bivariate regressions):

"We related seed rain of our most common exotic taxa to the abundance of those taxa, and life-history stages for *R. cathartica*, using ranged major axis (RMA) regression."

- No description of including additional covariates/adjustment terms (e.g., study area/plot, other vegetation structure, etc.) is provided alongside these analysis descriptions.

Reasoning (step-by-step)

1. The question asks whether the analysis took confounders into account appropriately (i.e., whether authors adjusted/controlled for confounding in the analysis).
2. The paper describes two main analytic approaches:
 - ANOVA comparing seed counts by **habitat, perch presence**, and their **interaction** (i.e., group comparisons across these design factors).
 - RMA regression relating **seed rain** to **abundance/density** of the focal taxa (and buckthorn life stages), which is described as relating one outcome to one predictor at a time.
3. In the quoted analysis descriptions, the authors do **not** state that they adjusted for other potential confounders (e.g., study area/block, plot-level differences, other habitat attributes, or any multivariable model including multiple confounders simultaneously).
4. Because the paper does not explicitly report any confounder-adjusted analysis beyond including habitat/perch as factors in ANOVA (and does not describe controlling for additional confounders in the regression analyses), it **strongly suggests** confounding was not comprehensively accounted for.

Final judgment

[[seemingly no]] === PARSED OUTPUT === seemingly no